

Project 8 - Part A

Erin Plaster

Texas A&M Central Texas
MATH-4309-110 Advanced Analysis
Dr. Mienie Scholtz
December 11, 2025

1 Problem Statement

The purpose of this project is to discuss Riemann sums and show how they can be used to approximate the area under a curve, including what conditions must be present for a function to be Riemann integrable. More specifically, the project will look closely at the convergence of upper and lower sums for a continuous function on a closed interval and discuss how integrability is affected when discontinuities or unbounded behavior are present. In this report, the function $f(x) = x^2$ is used to illustrate the convergence of Riemann sums when more and more partitions are made. The project also looks at other examples, such as $f(x) = \tan(x)$ and $f(x) = \frac{1}{x}$ to discuss when Riemann integration fails. The goal of the model is to formalize the mathematical structure that is required for Riemann integration and to demonstrate convergence through numerical or geometric methods. The project will also assess limitations for functions that have discontinuities or unbounded values. The requirements for the project included a GeoGebra applet that shows Riemann integrability using upper and lower sums and a short presentation discussing the model, reasoning, and results [uer25].

2 Mathematical Formulation

To calculate a Riemann sum, we consider a closed and bounded interval $[a, b]$ for a given function $f(x)$. We then divide the interval into subintervals, or partitions, and select a sample point within each partition to evaluate $f(x)$. The sum of the resulting rectangular areas approximates the total area under the curve of $f(x)$ over $[a, b]$. In Riemann integration, we also define upper and lower sums based on the supremum and infimum of $f(x)$ for each subinterval, and as the partitions become finer (tending towards zero), the upper and lower sums converge to the same value which represents the exact area under the curve $f(x)$.

2.1 Boundedness and Extrema

We should first consider what it means for a set, or in our case, a function, to have bounds. A function f defined on a closed interval $[a, b]$ must be bounded in order for lower and upper sums to exist. A set is considered to be bounded above if it has an upper bound, bounded below if it has a lower bound, and bounded only if both exist. The least upper bound (supremum) and the greatest lower bound (infimum) are required to define the heights of the rectangles used in Riemann sums. By definition, we can define the upper and lower bounds as:

A set $A \subset \mathbb{R}$ of real numbers is bounded from above if there exists a real number $M \in \mathbb{R}$, called an upper bound of A , such that $x \leq M$ for every $x \in A$. Similarly, A is bounded from below if there exists $m \in \mathbb{R}$, called a lower bound of A , such that $x \geq m$ for every $x \in A$. A set is bounded if it is bounded both from above and below. Equivalently, a set A is bounded if $A \subset I$ for some bounded interval $I = [m, M]$. [Hun14].

2.2 Partitions

Riemann integration requires partitions, or subintervals, defined as:

Let I be a nonempty, compact interval. A *partition* of I is a finite collection $\{I_1, I_2, \dots, I_n\}$ of almost disjoint, nonempty, compact subintervals whose union is I .

A partition of $[a, b]$ with subintervals $I_k = [x_{k-1}, x_k]$ is determined by the set of endpoints of the intervals $a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$.

We can denote a partition P either by its intervals $P = \{I_1, I_2, \dots, I_n\}$ or by the set of endpoints of the intervals $P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$. [Hun14]

These partitions can be divided to create smaller subintervals. Refinements allow us to compare upper and lower sums on increasingly finer partitions, which is essential to determine integrability. Refinements squeeze the upper and lower sums closer together, which give us better and better approximations for integration.

2.3 Upper and Lower Riemann Sums

The definitions of supremum and infimum are important concepts for integration, namely the Riemann integral.

Suppose that $A \in \mathbb{R}$ is a set of real numbers. If $M \in \mathbb{R}$ is an upper bound of A such that $M \leq M'$ for every upper bound M' of A , then M is called the least upper bound or *supremum* of A , denoted

$$M = \sup A.$$

If $m \in \mathbb{R}$ is a lower bound of A such that $m \geq m'$ for every lower bound m' of A , then m is called the greatest lower bound or *infimum* of A , denoted

$$m = \inf A.$$

If $A = x_i : i \in I$ is an indexed subset of $\mathbb{R}$, we also write

$$\sup A = \sup_{i \in I} x_i, \quad \inf A = \inf_{i \in I} x_i.$$

[Hun14]

This definition further indicates that the supremum and infimum of a function are necessarily unique (if they exist) and the supremum is greater than the infimum.

2.4 The Riemann Integral

The Riemann integral is the simplest form of integration which allows us to integrate continuous (and some discontinuous) functions over an interval, by using partitions to calculate the area under the curve. If integration is required for a highly discontinuous or more complex function, there are other options such as the Lebesgue integral.

The definitions of the Riemann sum are:

The upper Riemann sum of f with respect to the partition P by

$$U(f; P) = \sum_{k=1}^n M_k |I_k| = \sum_{k=1}^n M_k (x_k - x_{k-1})$$

and the lower Riemann sum of f with respect to the partition P by

$$L(f; P) = \sum_{k=1}^n m_k |I_k| = \sum_{k=1}^n m_k (x_k - x_{k-1})$$

[Hun14]

What this means is that we calculate the sum of each partition, using the supremum and infimum within each subinterval to find the area under the curve. As our partitions become finer (with widths approaching zero), the values of the upper and lower sums converge to the same limit. This common value gives the exact area under the curve for a given function $f(x)$, on the interval $[a, b]$, defining the Riemann integral. The formal definition of the Riemann Integral is:

A function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable on $[a, b]$ if it is bounded and its upper integral $U(f)$ and lower integral $L(f)$ are equal. In that case the Riemann integral f on $[a, b]$ denoted by

$$\int_a^b f(x) \, dx, \quad \int_a^b f, \quad \int_{[a,b]} f$$

or similar notations is the common value of $U(f)$ and $L(f)$. [Hun14]

3 Analytical / Numerical Solution Approach

This section describes how Riemann sums were implemented using the model (GeoGebra applet).

3.1 Continuous Functions - Modeling $f(x) = x^2$

To visualize convergence, the interval $[a, b]$ was subdivided into smaller partitions of different widths (the width and number of subintervals can be adjusted in the applet). For each partition:

1. The supremum and infimum of $f(x) = x^2$ were computed on each subinterval.
2. Upper and Lower sums were calculated using the definitions in Section 2.
3. A GeoGebra applet visualized the rectangles corresponding to M_k and m_k .
4. The partition width was reduced to show that the upper and lower sums converge to the same value.

3.2 Discontinuous or Unbounded Functions

1. The same method was used to look at discontinuous or unbounded functions, such as $f(x) = \tan(x)$ on $[0.2]$. Because $\tan(x)$ has vertical asymptotes at $x = \frac{\pi}{2}$, the supremum is $+\infty$ and the infimum is $-\infty$. As the partitions become finer, neither the upper or lower sums converge, so the function is not Riemann integrable on this interval. It is important to note, however, that if the interval were restricted to not include an asymptote, the function would be continuous and is then possible to integrate on the more restricted interval.
2. The suprema and infima were evaluated using GeoGebra to illustrate that the function is not integrable near discontinuities.

4 Interpretation of Results

4.1 Convergence for Continuous Functions

Figures 1 and 2 show the calculation for $f(x) = x^2$ with a smaller and larger partition number. We can see that as the number of partitions increase (each partition width decreases), the upper and lower sums steadily converge, confirming the theoretical condition for Riemann integrability. The narrowing gap between these sums demonstrates that the integral exists and equals the area under the curve (shown in purple).

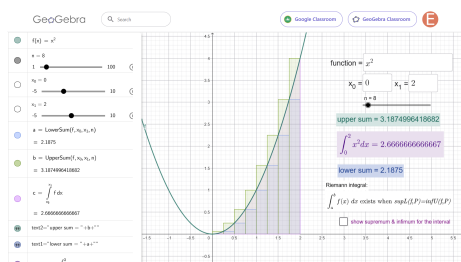


Figure 1: Visualizing a lower number of partitions for $f(x) = x^2$

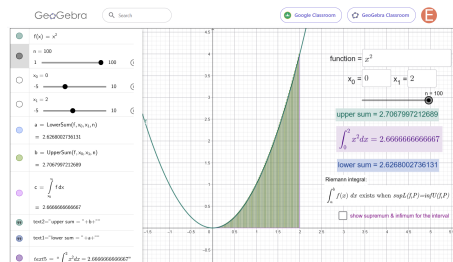


Figure 2: Visualizing a higher number of partitions for $f(x) = x^2$

4.2 Behavior of Non-Integrable Functions

For functions that have a vertical asymptote or unbounded intervals, such as $f(x) = \tan(x)$, the upper and lower sums do not converge, as mentioned above. This behavior follows the requirement that a function must be bounded on $[a, b]$ to be Riemann integrable. It is important to note that the applet can indicate incorrect infima because GeoGebra samples many values near the asymptote, and depending on where it samples most values, an unbounded function can have incorrect values given for the infimum and supremum. Any closed interval that avoids the discontinuity will be Riemann integrable on the interval because it will be continuous and bounded, and the upper and lower sums will converge to the same value.

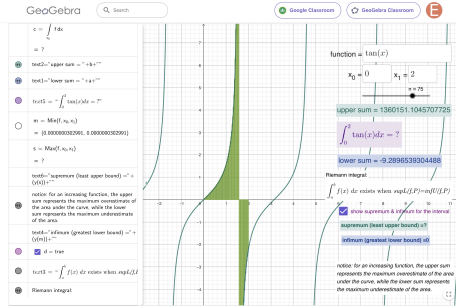


Figure 3: $f(x) = \tan(x)$ on $[0, 2]$

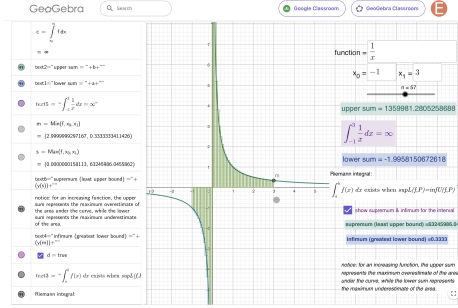


Figure 4: $f(x) = \frac{1}{x}$ is discontinuous on $[-1, 3]$

4.3 Visualization of Bounded Functions

Bounded functions are functions with boundaries or constraints, or functions that stay within fixed limits. For example, $f(x) = \sin(x)$, is bounded because its range is between $[-1, 1]$. An example of a function with only a lower bound is $f(x) = x^2$; its lowest point never drops below zero and grows infinitely as x increases. The function $f(x) = \frac{1}{x}$ is unbounded, as shown above.

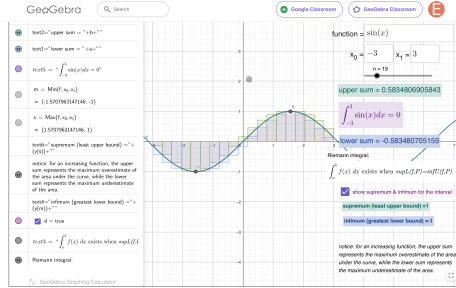


Figure 5: $f(x) = \sin(x)$ bounded within the range of $[-1, 1]$

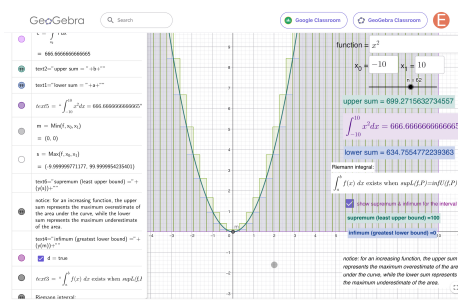


Figure 6: $f(x) = x^2$, lower bound only at 0.

5 Assumptions and Limitations

5.1 Assumptions

1. The functions examined are assumed to be bounded, unless otherwise noted.
2. The partition points are assumed to be approximate true suprema and infima within subintervals.
3. Numerical tools, such as GeoGebra, sample sufficiently close to approximate extrema.
4. The mathematical model assumes that the interval $[a, b]$ is closed and finite.

5.2 Limitations

1. Numerical software, such as GeoGebra, identifies suprema and infima through discrete sampling, which may miss extreme values near asymptotes or rapid oscillations. This may produce a falsely large/small suprema/infima where an asymptote actually exists.
2. Riemann sums converge slowly for functions with steep curvature, making them less practical than more advanced integration methods (such as the Lebesgue integral).
3. Highly discontinuous or unbounded functions can not be calculated using Riemann integrability because the upper or lower bound diverges.

6 Conclusion

This project demonstrated how Riemann sums can be used to model the area under a curve and to determine whether a function is integrable on a given interval. Continuous functions, such as $f(x) = x^2$ showed convergence of upper and lower sums, which confirms that they are Riemann integrable. In contrast, functions with vertical asymptotes or unbounded behavior did not converge, indicating that boundedness is important in Riemann integration. Visualizing the upper and lower sums helped determine when discontinuities or problems might exist when computing Riemann integration.

References

- [Hun14] John Hunter. *An Introduction to Real Analysis*. 2014. URL: https://www.math.ucdavis.edu/~hunter/intro_analysis_pdf/intro_analysis.pdf.
- [uer25] u/erinplaster. *Riemann Upper and Lower Sums*. 2025. URL: <https://www.geogebra.org/m/xhp3sqnt>.